Consider the following two statements:

S3: I had lunch yesterday with my college mate at Hotel Plaza.

S4: Donald Trump had a luncheon meeting with the German chancellor on the eve of the NATO summit.

Both S3 and S4 are factual statements. While both can be potential claims, S3 is about a commonplace event and does not involve any public entities or events that would be of interest to the public. Hence it is not necessary to verify its authenticity. On the other hand, S4 involves public entities. Since this information would be of interest to the public, as it involves politicians, it needs to be fact-checked.

It is important to note that claim detection involves two distinct processes and each of these has distinct linguistic characteristics. The first characteristic is that it should be objective or factual. The second characteristic is that the information contained in it should be of interest or use to the public, which is why it needs to be verified. While it is easy for humans to intuitively know whether or not a particular sentence could be of interest to the general public, how can we design machine learning based approaches for detecting this characteristic? This is an extremely challenging task from a machine comprehension perspective. For instance, consider the following two sentences:

S5: Donald Trump dodged the army draft when he was young.

S6: My brother John dodged the army draft by claiming disability, when he was young.

These two sentences are very similar in content. Both are also objective statements. However, since S5 is about the actions of a public individual, it passes the criterion of information of public interest, whereas S6 is not about any public individual and hence fails the public interest criterion. However, based on the textual content alone, it would be very difficult to design a machine learning based system, which can classify the first statement as a claim that would be of interest to the public while classifying the second statement as being of no interest to the public.

What are some of the features we can think of for detecting claims of interest to the public, which need to be checked? One possibility is to identify the 'Named Entities' in the claim sentence and try to determine whether they are check worthy. However this can be misleading in many cases. Consider the statement below:

S7: I watched the movie 'Mission Impossible' starring Tom Cruise.

A Named Entity Recognition System would return Tom Cruise as an entity, and it may be possible to use an external knowledge source such as Wikipedia to determine whether the recognised entity is a public figure or not. However, S7 is not a claim that needs to be fact-checked since it is about the actions of an individual, and not a public entity. Hence it is necessary to figure out whether the claim is about the actions or attributes of a public entity rather than just checking whether a mention of a public entity is present in the claim. This requires us to parse the sentence and use the dependency relations information as a feature.

By now, some of our readers would have realised that the general public can also be interested in information that is not about celebrities/public entities. For example, consider the following statement:

S8: There is a new cancer cure in Cuba which has been effective in more than 90 per cent of the cases.

This statement is not about any public entity; however, it is an objective statement and the information contained in it would be of interest to the public. Hence this statement surely needs to be fact-checked. So how do we characterise S8? It definitely contains information, which would be useful to the general public. Hence, though it may not have any celebrity related content, it has content that could be of great use to the public.

While we have discussed what makes a piece of information interesting to the general public, this characteristic of the 'interestingness' of information is extremely subjective and difficult to capture easily in terms of the linguistic and non-linguistic features of the text. Instead of having to explicitly identify the features that would allow us to classify claims which are of interest to the public from those that are not, this problem is better suited for deep learning approaches that do not require explicit feature extraction. In such a supervised deep learning approach, we provide as input to the network, the claim sentences that are labelled as interesting or not interesting to the public and let the network on its own identify the features that characterise each category, through back propagation. However, this approach requires that we have a sufficiently large set of sentences so that all various kinds of 'interestingness' can be covered by the training set. We will discuss such a neural network representation for classifying 'interesting' sentences in our next column.

If you have any favourite programming questions or software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Wishing all our readers happy coding until next month! END

By: Sandya Mannarswamy

The author is an expert in systems software and is currently working as a research scientist at Conduent Labs India (formerly Xerox India Research Centre). Her interests include compilers, programming languages, file systems and natural language processing. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group 'Computer Science Interview Training India' at *http://www.linkedin.com/groups?home=&gid=2339182*.